

■ Where Is My BMTC

Product Requirements Document

Version	1.0
Date	May 2026
Status	Draft
Platform	Flutter · FastAPI · MongoDB
Target City	Bengaluru, India

1. Executive Summary

Where Is My BMTC is a real-time transit tracking application for Bengaluru Metropolitan Transport Corporation (BMTC) buses. The product enables commuters to search routes, view stop sequences, and track buses on a live map — solving the daily uncertainty faced by millions of Bengaluru bus riders. The MVP will be delivered in phased versions starting with a static transit directory and evolving toward AI-powered transit intelligence.

2. Problem Statement

BMTC operates one of the largest urban bus networks in India, yet commuters have no reliable digital tool to:

- Know **which bus to take** from their current location
- See **real-time bus position** on a map
- Estimate **arrival times** at a stop
- Plan multi-leg journeys across the network

This results in long wait times, missed buses, and over-reliance on auto-rickshaws for distances easily covered by BMTC.

3. Goals & Success Metrics

Goal	Metric	Target
Route discovery	Search-to-result time	< 2 seconds
Live tracking accuracy	Bus position error	< 50 metres
User retention	D7 retention rate	> 30%
App reliability	Uptime	99.5%
Onboarding	Time to first search	< 60 seconds

4. Product Scope & Versioning

Version 1 — Static Transit App

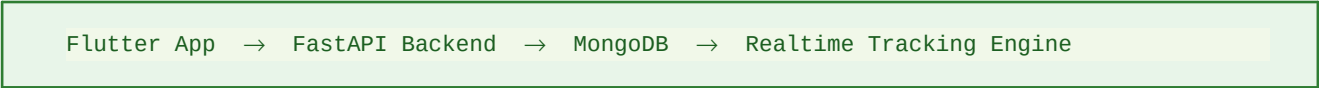
- Bus & route search by number or name
- Complete stop-sequence listing per route
- Static map with route polyline & stop markers
- Offline-capable dataset (buses, routes, stops JSON)

- Version 2 — Simulated Live Tracking**
 - Animated bus movement between stops (3-second intervals)
 - Real-time UI updates via WebSocket
 - Fake ETA calculation based on simulation speed
- Version 3 — Actual GPS Tracking**
 - Integration with live GPS source
 - FastAPI WebSocket → Flutter live map pipeline
 - MongoDB geo-indexed bus position storage
- Version 4 — ETA Prediction**
 - Historical travel-time ML model
 - Traffic-aware arrival estimates
 - Push notifications for approaching buses
- Version 5 — AI Transit Intelligence**
 - Natural-language journey planner
 - Multi-route optimization
 - Crowding prediction & alternative suggestions

5. Technical Architecture

Layer	Technology	Rationale
Frontend	Flutter	Cross-platform (iOS & Android), rich map widgets
Backend	FastAPI (Python)	Async, high-performance, auto-generated API docs, WebSocket support
Database	MongoDB	Flexible geo-queries, fast prototyping, realtime-friendly
Realtime	WebSockets	Low-latency push for live bus positions
Maps	Mapbox / Google Maps	Route polylines, stop markers, user location
Hosting	MongoDB Atlas + Cloud VM	Managed DB, scalable backend hosting

Architecture Flow



6. Core Data Models

6.1 buses.json

```
{
  "bus_number": "500D",
  "source": "Majestic",
  "destination": "Whitefield",
  "route_id": "R001"
}
```

6.2 stops.json

```
{
  "stop_name": "Majestic",
  "location": {
    "type": "Point",
    "coordinates": [77.572, 12.976]
  }
}
```

6.3 routes.json

```
{
  "route_id": "R001",
  "stops": ["Majestic", "Hebbal", "Marathahalli", "Whitefield"],
  "frequency_mins": 15,
  "operating_hours": "05:00-23:00"
}
```

7. API Specification (V1)

Method	Endpoint	Description	Response
GET	/	Health check	{"message": "Where is My BMTC API"}
GET	/buses	List all buses	Array of bus objects
GET	/search?query=500D	Search bus by number/name	Filtered bus list
GET	/bus/{bus_number}	Bus details + stops	Bus object with stop array
GET	/stops/nearby?lat=&lng=	Stops within 1 km	Array of stop objects
WS	/ws/track/{bus_id}	Live bus position stream	JSON position events

8. Feature Requirements

8.1 Home Screen

- Full-width search bar (search by bus number, route, or stop name)
- Recent searches persisted locally
- Quick-access tiles for popular routes

8.2 Bus Details Screen

- Source → Destination header with operating hours
- Scrollable stop-sequence list with ETA column (simulated in V2)
- 'View on Map' CTA button

8.3 Map Screen

- Route polyline rendered on Mapbox / Google Maps
- Stop markers with tap-to-see-name interaction
- Animated bus icon moving between stops (V2+)
- User location dot with 'Nearest Stops' radius ring

8.4 Live Tracking (V3+)

- WebSocket connection indicator (connected / reconnecting / offline)
- Bus position updated every 3 seconds
- Automatic reconnection on network loss

9. 7-Day MVP Build Plan

Day	Focus	Deliverable
Day 1	Project Setup	FastAPI + MongoDB + Flutter + GitHub repo initialised
Day 2	Datasets	buses.json, routes.json, stops.json populated
Day 3	Backend APIs	GET /buses, /search, /bus/{id} endpoints live
Day 4	Flutter Search UI	Home screen + Bus Details screen built
Day 5	Frontend ↔ Backend	Flutter consuming FastAPI over HTTP
Day 6	Map Integration	Route polyline + stop markers on Mapbox/Google Maps
Day 7	Simulated Tracking	Animated bus icon moving between stops

10. Out of Scope for MVP

✗ AI / machine-learning features

- ✗ Real BMTC API integration (unofficial APIs are unreliable)
- ✗ User authentication or accounts
- ✗ Admin / driver panels
- ✗ Complex realtime cloud scaling
- ✗ Ticket booking or payment flows

11. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
No official BMTC data feed	High	High	Seed app with manually curated JSON datasets; community contributions la
GPS hardware unavailable for V3	Medium	Medium	Simulate tracking in V2 to build the UI pipeline first
Map API cost overrun	Medium	Low	Use Mapbox free tier (50k loads/month) for MVP; defer billing to scale
Flutter map plugin instability	Low	Medium	Pin dependency versions; maintain fallback to Google Maps SDK

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